CE 310 — Week 11 Pre-Lab Quiz

Week: 11 | Topic: Hypothesis Testing — t-Tests and ANOVA Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers null/alternative hypotheses, p-values, Type I and II errors, the t-test, ANOVA, effect size, and the Python functions you will use in today's lab. Review the Week 11 Background reading before completing this quiz.

Questions

Q1. In a hypothesis test, the null hypothesis (H_0) is:

(A) The outcome you are hoping to prove with your data
(B) A statement of "no effect" or "no difference" that you are trying to disprove with evidence
(C) Always the hypothesis that the population mean equals zero
(D) Accepted as true only after the p-value exceeds 0.05

Q2. The p-value in a hypothesis test is best described as:

(A) The probability that H_0 is true
(B) The probability that H_1 is true
(C) The probability of observing a test statistic at least as extreme as the one computed, if H_0 were true
(D) The fraction of the sample that supports H_0

Q3. You compute a t-statistic of 3.5 with a corresponding p-value of 0.0005 and your significance level is $\alpha = 0.05$. You should:

(A) Fail to reject H_0 because the t-statistic is positive
(B) Reject H_0 because $p < \alpha$
(C) Fail to reject H_0 because you need at least 1,000 observations to trust a t-test
(D) Reject H_0 only if the sample size is greater than 30

Q4. A Type I error occurs when:

(A) You fail to reject H_0 even though H_1 is actually true
(B) You reject H_0 even though H_0 is actually true (a false positive)
(C) The test statistic is negative
(D) The sample size is too small to detect a real effect

Q5. The t-statistic in a two-sample t-test is approximately:

(A) The ratio of the sample variance to the population variance
(B) The product of each group's sample size
(C) The difference in group means divided by the standard error of that difference
(D) The square root of the F-statistic from a one-way ANOVA

Q6. A one-way ANOVA is the appropriate test when you want to:

(A) Determine whether a single sample mean equals a hypothesised value
(B) Compare exactly two group means
(C) Compare means from three or more groups simultaneously while controlling the overall Type I error rate
(D) Test whether two variables are linearly correlated

Q7. In the DOE Medium Office dataset, you run a two-sample t-test comparing mean Cooling_kWh for occupied hours (BH = 1) vs. unoccupied hours (BH = 0) and get $p \approx 0$. Your sample has 8,760 observations. Which conclusion is most defensible?

(A) The p-value of 0 proves H_0 is false with certainty
(B) The test is invalid because the two groups have very different standard deviations
(C) There is extremely strong statistical evidence that mean cooling differs between occupied and unoccupied hours
(D) The result is meaningless because large samples always produce small p-values

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Q8. Cohen's d is a measure of:

(A) The probability of rejecting H_0 when H_1 is true (statistical power)
(B) The standardised difference between two group means — how large the effect is in standard-deviation units
(C) The fraction of variance explained by a regression model
(D) The ratio of the F-statistic to the degrees of freedom in an ANOVA

Q9. A dataset has $n = 8{,}760$ observations. You compute a t-statistic of 2.1 with $p = 0.036$, and Cohen's $d = 0.05$. Which statement best describes this result?

(A) The effect is statistically significant and practically large

(B) The effect is statistically significant but practically very small ($d < 0.2$ is conventionally "negligible")
(C) The effect is not statistically significant because $d < 0.2$
(D) The result is uninterpretable without knowing the population standard deviation

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Q10. In Python, scipy.stats.f_oneway(group1, group2, group3) does which of the following?

(A) Fits a linear regression with three predictor variables
(B) Computes three separate t-tests, one for each pair of groups
(C) Runs a one-way ANOVA and returns the F-statistic and p-value for the hypothesis that all three group means are equal
(D) Returns the variance of each group separately
